

John Gargalionis

POSTDOCTORAL RESEARCHER · PARTICLE PHENOMENOLOGY

University of Valencia and IFIC

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Employment

University of Valencia

Valencia, Spain

POSTDOCTORAL RESEARCHER

Jan 2022

- Studies of lepton- and baryon-number violation
- Delayed due to pandemic

University of Melbourne

Melbourne, Australia

RESEARCH ASSISTANT

May – Oct 2021

- Developing novel techniques for quark–gluon discrimination using unsupervised machine learning and the CMS Open Data
- Extending doctoral work on connection between flavour anomalies and neutrino masses

Education

University of Melbourne

Melbourne, Australia

DOCTOR OF PHILOSOPHY (THEORETICAL PARTICLE PHYSICS)

2016 – 2020

- *Thesis title:* Models of radiative neutrino mass and lepton flavour non-universality
- *Supervisor:* Prof. Raymond VOLKAS

MASTER OF SCIENCE (WITH DISTINCTION)

2014 – 2016

- *Thesis title:* Neutrino mass through leptoquarks: a new radiative model and its experimental prospects
- *Supervisors:* Prof. Raymond VOLKAS and Prof. Elisabetta BARBERIO

BACHELOR OF SCIENCE

2010 – 2014

- *Specialisations:* Physics, Neuroscience, Ancient Languages

Publications

Research publications

EXPLODING OPERATORS FOR MAJORANA NEUTRINO MASSES

2021

John Gargalionis & Raymond R. Volkas

JHEP

arXiv:2009.13537, JHEP 01 (2021)

RADIATIVE NEUTRINO MASS MODEL FROM A MASS DIMENSION 11 $\Delta L = 2$ EFFECTIVE OPERATOR

2020

John Gargalionis, Iulia Popa-Mateiu & Raymond R. Volkas

JHEP

arXiv:1912.12386, JHEP 03 (2020)

A NEAR-MINIMAL LEPTOQUARK MODEL FOR RECONCILING FLAVOUR ANOMALIES AND GENERATING RADIATIVE NEUTRINO MASS

2019

Innes Bigaran, John Gargalionis & Raymond R. Volkas

JHEP

arXiv:1906.01870, JHEP 10 (2019)

RECONSIDERING THE ONE LEPTOQUARK SOLUTION: FLAVOUR ANOMALIES AND NEUTRINO MASS

2017

Yi Cai, John Gargalionis, Michael A. Schmidt & Raymond R. Volkas

JHEP

arXiv:1704.05849, JHEP 10 (2017)

EXPLAINING THE 750 GeV DIPHOTON EXCESS WITH A COLOURED SCALAR CHARGED UNDER A NEW CONFINING GAUGE INTERACTION

2016

Robert Foot & John Gargalionis

PRD

arXiv:1604.06180, Phys. Rev. D94:011703 (2016)

Other publications

SOLUTIONS TO PROBLEMS AT LES HOUCHES SUMMER SCHOOL ON EFT

2020

Marcel Balsiger, Marios Bounakis, Medhi Drissi, John Gargalionis, Erik Gustafson, Greg Jackson, Matthew Leak, Christopher Lepenik, Scott Melville, Daniel Moreno, Michele Tommaro, Selim Touati & Timothy Trott

Oxford University Press
(included in publication of
lecture notes)

arXiv:2005.08573

General Skills

Programming

Python (SciPy, NumPy, SymPy, Jupyter, etc.), Mathematica (Wolfram Language), C/C++, Clojure(Script), Scheme, Haskell

- Practised in test-driven software development and version control using git
- Experience working with high-performance computing clusters using PBS and slurm

Data analysis

ROOT, SciPy, Pandas, Dask, h5py, SQLite

Machine learning

TensorFlow, Keras, (Un)supervised learning, Image classification

Communication

Public speaking, scientific writing, science communication

Academic Skills

Model building and model exploration

Radiative neutrino mass, group theory, flavour phenomenology, GUTs

- Constructing and analysing extensions of the Standard Model, experience mostly in models of radiative neutrino mass
- Expertise in the group theory of $SU(N)$ and $SO(N)$ in the context of GUT model building
- Knowledgeable about new-physics explanations of the B -meson anomalies and their connection to other topics in flavour
- Confident in analysing the viability of BSM models, especially in the context of flavour and collider constraints

Effective Field Theory

SMEFT, WET, $\Delta L = 2$ EFT

- Worked extensively with the Standard Model EFT at mass-dimension 6 and higher
- On top of current developments in operator-basis generation (i.e. Hilbert series and computational techniques)

Collider phenomenology

ROOT, CMSSW, MadGraph, Pythia, Fastjet, Delphes

- Confident in the generation of Monte-Carlo data starting from a Lagrangian
- Experience with the CMS analysis framework through the use of their publicly available data
- Well acquainted with concepts in experimental analysis, experience with a range of physics objects and signatures

Calculation and fitting

LieArt, Package-X, FeynRules, FeynArts, FormCalc, LoopTools, Flavio

- Calculating loop-level amplitudes by hand and with Mathematica packages (both analytically and numerically)
- Confident with the use of Flavio for flavour calculations and fits

Invited Seminars and Short Stays

May 2022 **Laboratory of Subatomic Physics & Cosmology (LPSC)**, Two-week research visit

Grenoble, France

Oct 2018 **Belle II Theory Interface Platform**, Leptoquarks and flavour

KEK, Japan

May 2018 **Monash University**, Radiative neutrino mass and the flavour anomalies: a circumstantial case

Melbourne, Australia

May 2017 **Instant workshop on B-meson anomalies**, Reconsidering the ‘one leptoquark’ solution: flavour anomalies and neutrino mass, <https://cds.cern.ch/record/2265323>

CERN, Switzerland

Contributed Seminars

Sep 2022	Korean Institute for Advanced Study (KIAS) , Exploding Operators for Majorana neutrino masses	Seoul, Korea
June 2022	HEFT 2022 , Exploding Operators for Majorana neutrino masses	Granada, Spain
May 2022	Instituto de Física Corpuscular (IFIC) , Exploding Operators for Majorana neutrino masses	Valencia, Spain
Nov 2020	PhD Completion Seminar , Models of radiative neutrino mass and lepton-flavour non-universality	Melbourne, Australia
Sep 2017	Geoff Opat Seminar Series , Radiative neutrino mass and the flavour anomalies: a circumstantial case	Melbourne, Australia
Aug 2017	Technische Universität Dortmund , Radiative neutrino mass and the flavour anomalies: a circumstantial case	Dortmund, Germany
Aug 2017	Technische Universität München , Radiative neutrino mass and the flavour anomalies: a circumstantial case	Garching, Germany
Dec 2016	APPC-AIP Congress , Reconsidering the 'one leptoquark' solution: flavour anomalies and neutrino mass	Brisbane, Australia
Jun 2016	University of Melbourne , Light leptoquarks at the LHC: neutrino mass and flavour physics	Melbourne, Australia
Nov 2015	MSc Completion Seminar , Radiative neutrino mass through leptoquarks	Melbourne, Australia

Training

Summer schools

Aug 2017	Joint Challenges for Cosmology and Colliders , MITP	Mainz, Germany
Jul 2017	EFT in Particle Physics and Cosmology , Ecole de Physique des Houches	Les Houches, France
Jul 2016	Pre-SUSY School , University of Melbourne	Melbourne, Australia

Summer research projects

Dark matter and heavy-flavoured quarks

ATLAS EXOTICS GROUP

- *Supervisor:* Dr. Francesca UNGARO

Melbourne, Australia

Jan 2016

Awards

Fellowships

Juan de la Cierva Fellowship

INSTITUTO DE FÍSICA CORPUSCULAR (IFIC)

- Total score: 90.10%
- Position declined

Valencia, Spain

2022

Scholarships & Honours

2021	Chancellor's prize , University of Melbourne	Melbourne, Australia
2018	Science Abroad Travel Scholarship , University of Melbourne	Melbourne, Australia
2017	Research poster award , CoEPP	Adelaide, Australia
2016 – 2019	Australian Postgraduate Award (APA) , Australian Research Council	Melbourne, Australia
2015	Prof. Kernot Research Scholarship in Physics , University of Melbourne	Melbourne, Austria
2014	N. D. Goldsworthy Scholarship , University of Melbourne	Melbourne, Austria

Teaching and Supervision

2022	Masters Supervision	<i>University of Valencia</i>
2020, 2021	Teaching assistant , Quantum Field Theory	<i>University of Melbourne</i>
2020	High School Teacher , VCE Modern Greek	<i>Victorian School of Languages</i>
2018	Tutor , 1st year Physics	<i>University of Melbourne</i>
2018, 2019	Tutor , 3rd year Quantum Mechanics	<i>University of Melbourne</i>
2018	Tutor , Advanced Scientific Programming in Python (Asia-Pacific)	<i>Melbourne Bioinformatics</i>
2016 – 2019	Teaching assistant and course content creator , 3rd year Subatomic Physics	<i>University of Melbourne</i>
2014 – 2016	Curriculum designer and language teacher , Modern and Ancient Greek	<i>Centre for Adult Education</i>
2014 – 2018	Laboratory demonstrator , 1st year Physics Lab, 3rd year Particle Lab, 2nd year Computational Physics	<i>University of Melbourne</i>
2010 – 2019	Private Tutor , Physics, Mathematics, Greek	

Scientific Outreach

2018, 2019	CoEPP Work Experience Program , Introductory talk: The Standard Model	<i>University of Melbourne</i>
2017	Physics Workshops , Project coordinator	<i>Hume Central Secondary</i>
2016	Undergraduate Seminar Series , Research talk targeted at undergraduate physics students	<i>University of Melbourne</i>
2015, 2016	International Masterclass in Particle Physics	<i>South Oakleigh Grammar</i>
2015	CoEPP Work Experience Program , Organising committee	<i>University of Melbourne</i>
2015	International Masterclass in Particle Physics	<i>University of Melbourne</i>

Other

Citizenship	Australian
Birth year	1991
Languages	English (native), Greek (C1), Spanish (B1)